Introduction to APIs: Wikipedia

Week 10 · APIs module · Textbook Chapter 10

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This week at a glance

We leave scraping behind and start asking servers for data through documented APIs.

Monday | Concepts & notebook intro

- What an API is, why it beats scraping, and why we learn on Wikipedia

Wednesday | Live-coding lab

- MediaWiki API: revisions, pageviews, links, raw requests
- We write the code together in `labs/lab-10.ipynb`

Friday | Continue the lab

- Finish the lab; check it against `labs/lab-10-key.ipynb`
- Final project proposal Q&A

Skipping weekly submissions

No lab notebook or GitHub submissions this week

Due Sunday, Oct 25

Final Project proposal (5%) — a research question, your data source(s) and method, and an ethics & feasibility check.

1. Monday · Concepts

Daily note

- **Example project ideas?** — Check the "Additional Exercises" sections in each textbook chapters for inspiration
- **Overly ambitious projects?** — Expensive, long, or large data retrieval tasks (proof of concept first!); advanced modeling or stats (descriptive summaries fine!)
- **How concrete of a RQ for final project?** — Not a contract, ok to change before end of course. Graded on 3 elements in assignment
- **Making Fridays more engaging?** — Experiment with extending Wednesday lab live-coding into Fridays
- **Final project overlap with class examples?** — Definitely!
- **Running Python at home?** — Everyone is different, but I still prefer to prototype and debug inside notebooks before translating into production scripts/pipelines

From scraping to APIs

For six weeks we **reverse-engineered** pages built for human eyes — inspecting HTML, rickety archives, guessing selectors, and cleaning up PDFs.

This week begins Part III. Instead of converting web content into data, we ask for the data publishers provide.

The skills you build here — reading documentation, constructing parameterized URLs, handling pagination, parsing nested JSON — transfer to every API in the rest of the course.

Wikipedia API already showed up throughout the course but now we will explore more of what we can do with it

What is an API?

An **API** (Application Programming Interface) is a **contract**. It specifies what you can request, which parameters you can send, and the format of the response.

- Clean **JSON**, not cluttered HTML
- **Stable endpoints**, not fragile CSS selectors (mostly)
- **Documentation** tells you exactly what exists and how to get it

The tradeoff: an API only exposes what its operator *chooses* to. You cannot request a field that the API does not include, exceed the rate limits, or rely on the API forever.

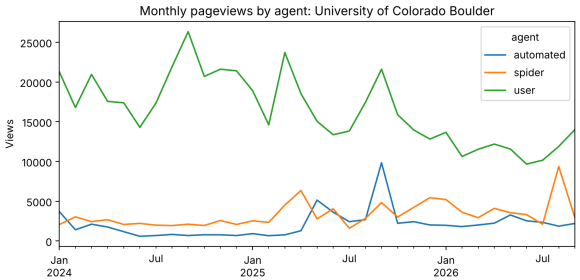
The "post-API age" from Week 03 should land harder now: many APIs have been restricted, repriced, or shut down — which is frustrating when you see how powerful they can be!

Why Wikipedia, and why now

One of the last big **open** data sources on the web: a free API, an open license, and public database dumps.

This openness is under pressure from AI:

- **Crawlers.** Bots made about 35% of pageviews but at least 65% of the costliest traffic
- **Readers.** With better bot detection, human pageviews for March–August 2025 came out about 8% lower than a year before.
- **Writing.** Editors clean up AI-written text, and a 2026 guideline bars using language models to write articles.



Monthly pageviews of the CU Boulder article, by who is reading.

A tour: what sits behind every article

Like most users, you have probably never clicked any deeper into Wikipedia than an article. But Wikipedia keeps many important records publicly available:

- who changed what, and when — the **revision history**
- what editors argued about — the **talk page**
- how many people read it — **pageviews**
- what it links to, what links to it, and the same topic in other **languages**
- its **references**, **categories**, and **templates**
- the editors themselves — **user pages** and **contributions**

Each one has an **API endpoint**. Anything you can see in the interface, you can count.

MediaWiki: the software behind Wikipedia

Wikipedia runs on **MediaWiki**, free and open-source wiki software. The same software runs Wiktionary, Wikidata, Wikimedia Commons, and thousands of other wikis.

- Every page you read is drawn from MediaWiki's database: the article's wikitext, its history, its links.
- The **Action API** serves that same database as data. Wikipedia's own tools use it, and so do the bots that patrol edits and researchers counting what's there.
- Every MediaWiki site has one, at `api.php`: learn it on Wikipedia, and it works on the others.

A basic request

```
https://en.wikipedia.org/w/api.php
?action=query                                  what to do
&titles=University_of_Colorado_Boulder         which page
&prop=info                                     what about it
&format=json                                   in what form
```

The "API call" boxes on the tour that follows show only what changes: the `prop` or `list` after `action=query`.

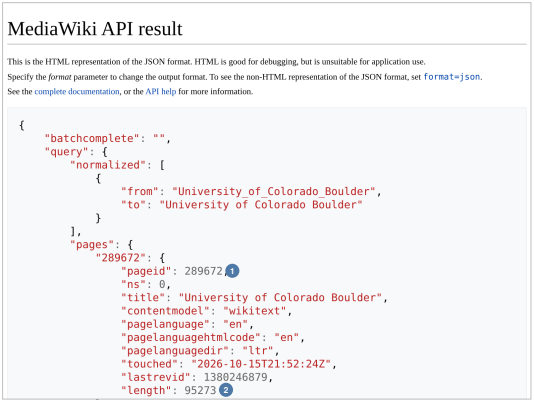
Documentation: mediawiki.org/wiki/API:Action_API

The anatomy of a request

Every Action API request is four parts:

1. a **base URL** — `.../w/api.php`
2. an `action` — what you want to do
3. **parameters** that narrow the request
4. a `format` — `json`

The response is nested JSON you navigate **one level at a time**:
`data["query"]["pages"][page_id]`.



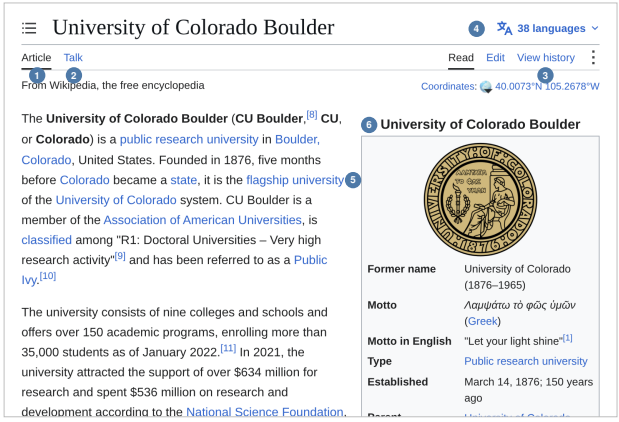
The same request in a browser, with `format=jsonfm`: the page ID (1) is the key under `pages`, and `length` (2) is the article's size in bytes.

From the interface to the API

What you see	Where	API request
The article's basic facts	the article	`prop=info`
Wikilinks out	the article's text	`prop=links`
What links here	Tools → What links here	`list=backlinks`
Templates	the infobox and other boxes	`prop=templates`
Categories	the foot of the article	`prop=categories`
References' links	References	`prop=extlinks`
Other languages	the languages button	`prop=langlinks`
Talk page	the Talk tab	any of these, for `titles=Talk:...`
Revision history	View history	`prop=revisions`
Pageviews	Page information	REST `/metrics/pageviews/...`
User contributions	Special:Contributions	`list=usercontribs`

Every Action API request here starts
`https://en.wikipedia.org/w/api.php?action=query&format=json`

The article



Tabs lead to the Article (1), its Talk page (2), and every past version (3). The languages button (4) lists other editions. A **wikilink** (5) points to another article; the **infobox** (6) comes from a **template**.

The API call

`action=query&prop=info` — the page's ID, length, and last edit

For research: Article length and last edit, across many articles: which topics Wikipedia covers in depth, and which it neglects.

Wikilinks, and what links here



A wikilink is a citation of another article. **What links here** is the reverse map: every page that links to this one (1), with filters (2).

The API call

Links out: `prop=links`
Links in: `list=backlinks`

For research: Link networks: what a topic connects to, and how central it is, measured by the pages that point to it.

Interlanguage links



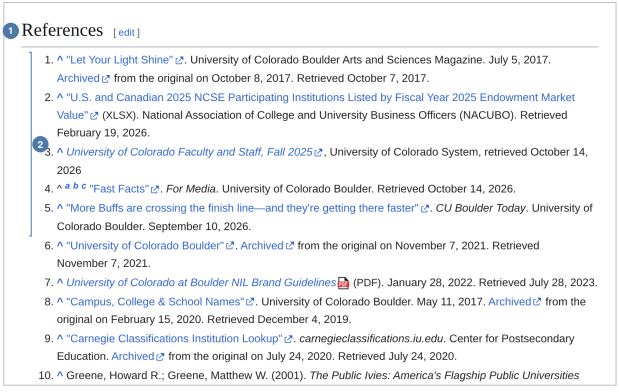
The same topic in other editions (2), searchable (1). Each edition is written by its own community — they are not translations.

The API call

```
prop=langlinks
```

For research: Global attention: which topics exist in which languages, and where coverage is missing.

References



1 References [edit]

1. ^ "Let Your Light Shine". University of Colorado Boulder Arts and Sciences Magazine. July 5, 2017. Archived from the original on October 8, 2017. Retrieved October 7, 2017.
2. ^ "U.S. and Canadian 2025 NCSE Participating Institutions Listed by Fiscal Year 2025 Endowment Market Value" (XLSX). National Association of College and University Business Officers (NACUBO). Retrieved February 19, 2026.
3. ^ *University of Colorado Faculty and Staff, Fall 2025*, University of Colorado System, retrieved October 14, 2026
4. ^ a b c "Fast Facts". *For Media*. University of Colorado Boulder. Retrieved October 14, 2026.
5. ^ "More Buffs are crossing the finish line—and they're getting there faster". *CU Boulder Today*. University of Colorado Boulder. September 10, 2026.
6. ^ "University of Colorado Boulder". Archived from the original on November 7, 2021. Retrieved November 7, 2021.
7. ^ *University of Colorado at Boulder NIL Brand Guidelines* (PDF). January 28, 2022. Retrieved July 28, 2023.
8. ^ "Campus, College & School Names". University of Colorado Boulder. May 11, 2017. Archived from the original on February 15, 2020. Retrieved December 4, 2019.
9. ^ "Carnegie Classifications Institution Lookup". *carnegieclassifications.iu.edu*. Center for Postsecondary Education. Archived from the original on July 24, 2020. Retrieved July 24, 2020.
10. ^ Greene, Howard R.; Greene, Matthew W. (2001). *The Public Ivies: America's Flagship Public Universities*

2

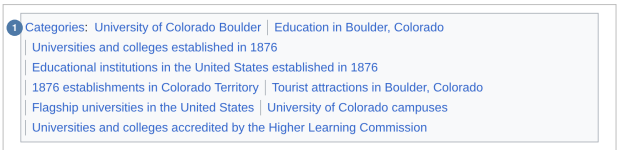
An article's claims rest on its references (1): numbered citations (2), many with an archived copy (Week 3).

The API call

`prop=extlinks` — every external link on the page

For research: Sourcing: which outlets Wikipedia relies on, and how many of its links still work or point to archives.

Categories and templates



Categories (1) file the article with others like it. Like the infobox's template, they are kept by editors — as consistent as the people who keep them.

The API call

```
prop=categories
prop=templates
```

For research: Sampling frames: every article in a category, such as all U.S. flagship universities, to compare.

The talk page



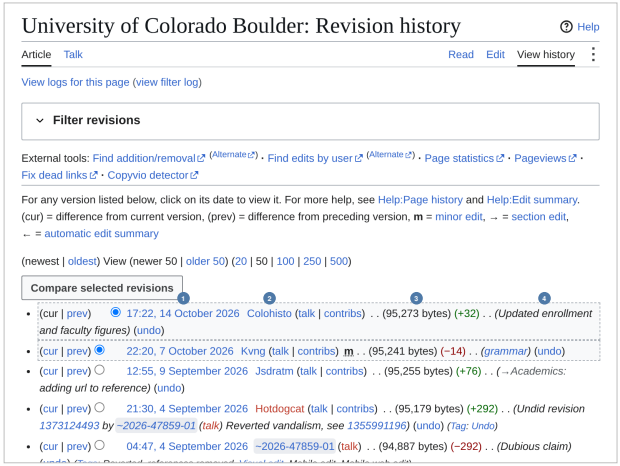
Where editors discuss changes (1). Its banners (2) record the article's quality rating and the WikiProjects that look after it.

The API call

Any of the same calls, for `titles=Talk:...`

For research: Deliberation: how editors argue, and how disputes are settled.

The revision history



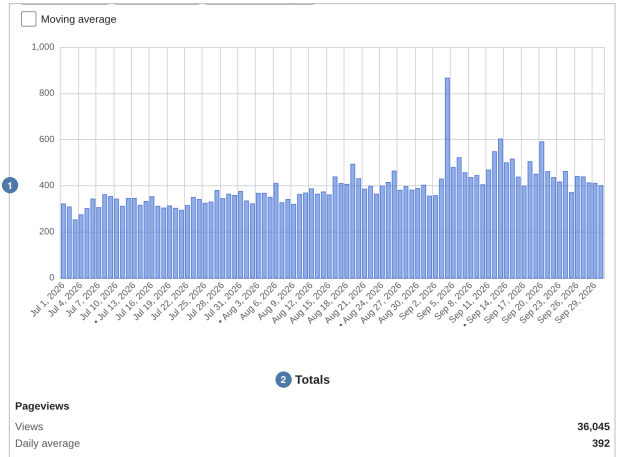
Every edit ever made, one row each: when (1), by whom (2), how many bytes (3), and the editor's summary (4).

The API call

`prop=revisions` — with `rvstart` to fix the window

For research: Collaboration over time: who edits, how much, and how fast an article changes when news breaks.

Pageviews



Wikipedia counts its readers and publishes the counts: daily views (1) and the period's total (2), here for July–September 2026.

The API call

A second API (REST):
`/metrics/pageviews/per-article/...`

For research: Public attention: what people look up, and when; spikes follow news and events.

User pages and contributions


User contributions for Jimbo Wales

Help

Results for Jimbo Wales (talk | block log | uploads | logs | global block log | global account | filter log)
A user with 15,133 edits. Account created on 27 March 2001.

Search for contributions

(newest | oldest) View (newer 50 | older 50) (20 | 50 | 100 | 250 | 500)

1. 12:19, 30 September 2026 (diff | hist) . . (+377) . . User talk:Jimbo Wales (→New cross-language page link: Shotgun: Reply) (Tag: Reply)
   3 2 4
- 19:50, 28 September 2026 (diff | hist) . . (+143) . . User talk:Jimbo Wales (→New cross-language page link: Shotgun: Reply) (Tag: Reply)
- 09:43, 27 September 2026 (diff | hist) . . (+496) . . User talk:Jimbo Wales (→Question about Deletionism and inclusionism: Reply) (Tag: Reply)
- 16:35, 25 September 2026 (diff | hist) . . (+218) . . User talk:Jimbo Wales (→Off Topic Question: Reply) (Tag: Reply)
- 16:35, 25 September 2026 (diff | hist) . . **(+550)** . . User talk:Jimbo Wales (→Question about using AI: Reply) (Tag: Reply)
- 18:50, 19 September 2026 (diff | hist) . . **(+603)** . . User talk:Jimbo Wales (→Update on Staff Union Vote and information from WMF: Reply) (Tag: Reply)
- 18:47, 19 September 2026 (diff | hist) . . (+385) . . User talk:Jimbo Wales (→Update on Staff Union Vote and information from WMF: Reply) (Tag: Reply)
- 21:56, 18 September 2026 (diff | hist) . . **(+1,034)** . . User talk:Jimbo Wales (→Update on Staff Union Vote and information from WMF: Reply) (Tag: Reply)
- 21:51, 18 September 2026 (diff | hist) . . **(+2,210)** . . User talk:Jimbo Wales (→Update on Staff Union Vote and

Editors have pages too. Jimbo Wales's contributions, across all of Wikipedia: when (1), which page (2), how many bytes (3), and the summary (4).

The API call

`list=usercontribs`
(the user page itself: `titles=User:Jimbo_Wales`)

For research: Editors' careers: what else an article's editors work on, and the communities they form.

Your Final Project proposal, due Sunday

Three parts, worth **5%** of your grade:

1. **A research question.** What the data could answer, and who would care about the answer. A question, not a topic: not "Wikipedia," but "Which events send readers to articles about wildfires?"
2. **Data source(s) and method.** The site, API, or documents, and how you'll get them: an API, scraping, PDFs, or a mix. Name the endpoint, page, or file.
3. **Ethics and feasibility.** The terms of service, `robots.txt`, and any personal data in it — and proof that you can get **one record**: paste the request and what came back.

Need inspiration?

Every chapter has *Additional Exercises* that could make great project questions.

Collecting over time?

Start the collector now, and say so in the proposal.

2. Wednesday · Live-Coding Lab

New this week: we live-code the lab together

Find it | `labs/` in the course repository

1. Open the **labs** folder of the course repository, `cuinfoscience/INF04617-Fall2026`, on GitHub
2. Click `lab-10.ipynb`, then **Download raw file**, as with the companion notebooks
3. Save it in your course folder, and open it in JupyterLab in the `webdata` environment

Use it | fill in the cells as we go

- Each step's text explains it; its code cell holds only a comment saying what to write
- I write the code on screen, and you write it in your copy
- Put **your** email in the User-Agent in Step 0
- Eleven steps, about 75 minutes: today, and Friday if we run over

Solutions

`labs/lab-10-key.ipynb`: every cell filled in and run on October 20. Fall behind? Copy the cell from the key and keep going.

Nothing to submit

The lab is for practice in class. The next frames show the same code, for reference.

A 429 is not a bug

It's Wikimedia's rate limit. The lab's `get()` waits the seconds it asks for, and tries again.

The four parts, then navigate the JSON

Assemble the request as a dictionary of parameters; `requests` builds the URL for you:

```
import requests
url = "https://en.wikipedia.org/w/api.php"
CLIENT = "WebDataScience/1.0 (your-email@colorado.edu)" # name/version (contact)
HEADERS = {"User-Agent": f"{CLIENT} python-requests/{requests.__version__}"}

params = {"action": "query", # what to do
          "titles": "University of Colorado Boulder", # about this page
          "prop": "info", # what we want back
          "format": "json"} # how we want it
data = requests.get(url, params=params, headers=HEADERS).json()
```

Then walk the nested response one key at a time — page IDs are the keys:

```
pages = data["query"]["pages"]
page_id = list(pages.keys())[0]
print(pages[page_id]["title"], pages[page_id]["pageid"], pages[page_id]["length"])
```

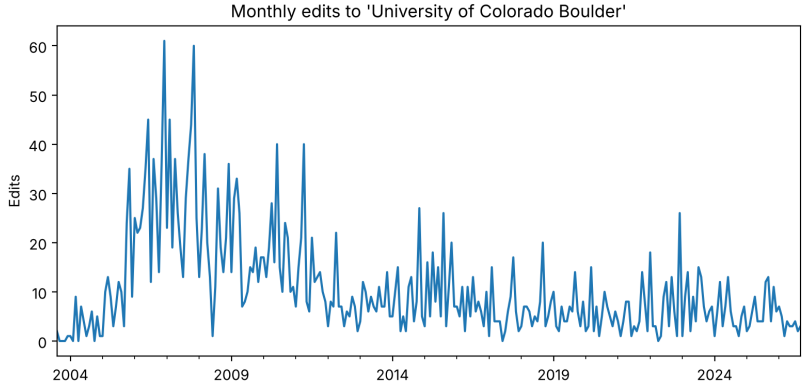
Revisions — and the pagination loop

```python
def get_revisions(title, until="2026-09-30T23:59:59Z"):
    params = {"action": "query", "titles": title,
        "prop": "revisions", "rvlimit": "max",
        "rvprop": "ids|timestamp|user|size|comment",
        "rvstart": until, # newest edit to include
        "format": "json"}
    all_revisions = []
    while True: # the pagination loop
        response = requests.get(url, params=params,
                                headers=HEADERS)
        response.raise_for_status()
        data = response.json()
        page = list(data["query"]["pages"].values())[0]
        all_revisions.extend(page.get("revisions", []))
        if "continue" in data:
            params.update(data["continue"]) # next batch
        else:
            break
    return pd.DataFrame(all_revisions)
```

The most common API bug: **forgetting to paginate**. The server returns 500 at a time and a `continue` token when more remain. Skip the loop and you silently keep a *fraction*.

`rvstart` fixes the window, so next week's run gives today's answer.

Every edit since 2003



2,988 revisions through September 2026; the ten most active editors made 15.9% of them. The busiest years came early, when the article was being written; since then, a few edits a month keep it current.

Pageviews: a second API

The REST API puts its parameters in the URL's *path*. Define the function once:

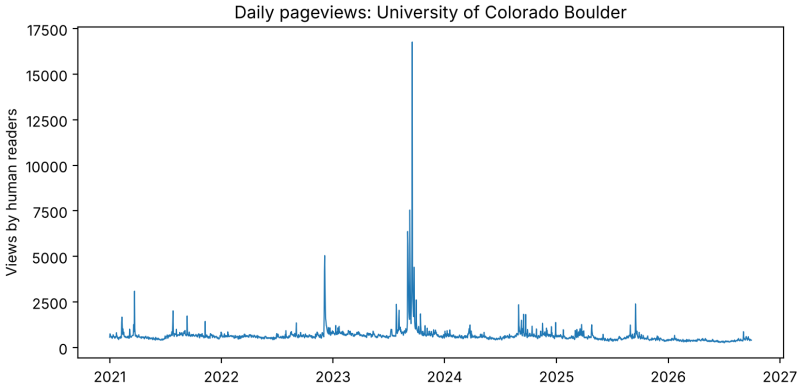
```
def get_pageviews(title, start, end,
                  agent="user", granularity="daily"):
    article = title.replace(" ", "_")
    url = ("https://wikimedia.org/api/rest_v1/metrics"
           "/pageviews/per-article/en.wikipedia"
           f"/all-access/{agent}/{article}"
           f"/{granularity}/{start}/{end}")
    response = requests.get(url, headers=HEADERS)
    response.raise_for_status()
    views = pd.DataFrame(response.json()["items"])
    views["timestamp"] = pd.to_datetime(
        views["timestamp"], format="%Y%m%d%H")
    return views[["timestamp", "views"]]
```

Define the function in one cell, and use it in the next:

```
cu_views = get_pageviews(
    "University of Colorado Boulder",
    "20210101", "20260930")
```

No `continue` here: one request returns the whole range.

Spikes are events

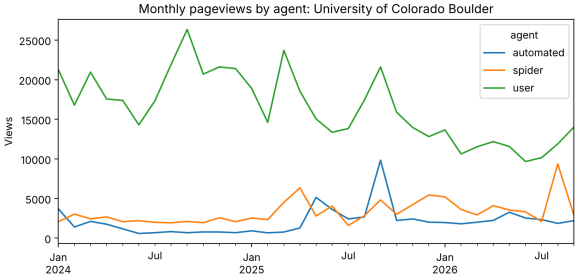


A typical day: about 600 human readers. December 5, 2022, two days after Deion Sanders was hired: 5,042. September 17, 2023, after a double-overtime win over Colorado State that ended late on the 16th: 16,767 — the API's days are **UTC** days.

What a pageview counts

A pageview is **not a person**: one reader who reloads three times is three views.

- The REST API's counts start in **July 2015**, under a definition drafted in 2014–15 (Research:Page view).
- Before that, pagecounts-raw (2007–2016) counted every request, edits and crawlers included. A fallback, not comparable.
- The **agent** says who is reading: `user` (people), `spider` (crawlers that say so), `automated` (bots that don't).
- When bot detection improves, past counts can be **reclassified** — as March–August 2025 were.



The same function, called once for each agent.

One function for every other feature

Links, categories, templates, languages, backlinks, and contributions are all `action=query` with a different `prop` or `list`. Write the continuation loop once:

```
def query_all(params):
    params = {"action": "query", "format": "json", **params}
    results = []
    while True:
        response = requests.get(url, params=params, headers=HEADERS)
        response.raise_for_status()
        data = response.json()
        results.append(data["query"])
        if "continue" in data:
            params.update(data["continue"])
        else:
            break
    return results

# page_list() runs query_all() for one page and one prop, and returns the items
categories = page_list(title, "categories", "cllimit")
backlinks = query_all({"list": "backlinks", "bltitle": title, "bllimit": "max"})
```

From raw requests to a client: mwclient

```
%conda install -y -c conda-forge mwclient

import mwclient
print(mwclient.__version__) # 0.10.1 from conda-forge

site = mwclient.Site("en.wikipedia.org",
    clients_useragent=CLIENT) # it adds mwclient/0.10.1
page = site.pages["University of Colorado Boulder"]

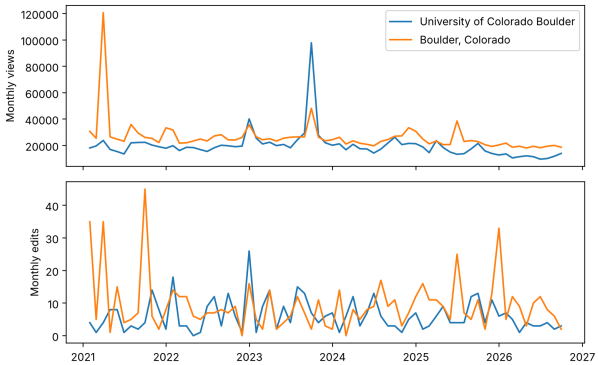
print(len(list(page.categories())))
print(len(list(page.langlinks())))
revisions = list(page.revisions( # 0.11: api_chunk_size=500
    start="2026-09-30T23:59:59Z", limit=500))
```

Each line replaces one of our functions — continuation included — and mwclient backs off when the servers report lag. A **429** still stops it: wait, and run it again.

Two things stay raw: **pageviews** (mwclient speaks the Action API only), and any parameter the library doesn't expose.

mwclient docs

Combining endpoints: the university and the city



The same functions, on **Boulder, Colorado**: monthly views (top) and edits (bottom), side by side.

Do articles get edited more when more people read them? Which events move the university's article, and which move the city's?

Finished the lab? Try these, then show-and-tell

Work in pairs. Do these exercises from the chapter, reusing the lab's functions:

1. Compare the revision histories of two related articles. Report the unique editors, the edits for each editor, and the monthly activity.
2. Get the daily pageviews of five articles. Report if the spikes occur together or separately.
3. Find the outgoing wikilinks of three related articles. Report the links that they share.
4. Read the API documentation. Write a request for a new endpoint such as `extlinks` or `recentchanges`.
5. Find the 20 most frequent editors of an article. Then find the other articles that they edit.
6. **5617**: reproduce one measure from Mesgari et al. (2015). For example, calculate a Gini coefficient of the edits for each editor

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

Error flavors

A **429** is Wikimedia's rate limit, which a whole classroom on one network can reach together: wait the `retry-after` seconds and run the cell again. No GitHub issue. A cell that fails the same way after waiting may be *genuinely* broken: confirm it, then search and file an issue.

3. Friday · Live-Coding Lab Continued

Today

First | finish the lab

- We pick up `labs/lab-10.ipynb` where we stopped on Wednesday
- Check your outputs against `labs/lab-10-key.ipynb`
- No notebook or GitHub submission this week

Then | proposal Q&A

- Bring your question and your source
- **Proposal due Sunday, October 25**

The Final Project: grade and dates

A **portfolio piece** that matches a web data source with a research design. Individually, or in pairs with the instructor's permission. **40%** of your grade.

Part	Due	Weight
Proposal	Sunday, October 25	5%
5-min presentation	Week 16, Nov. 30–Dec. 4	10%
Write-up	Monday, December 7	25%

What to expect

The proposal (Sunday, October 25)

- **A research question**: what the data could answer, and who would care about the answer.
- **Data source(s) and method**: the site, API, or documents, and whether you'll use an API, scraping, PDFs, or a mix.
- **Ethics and feasibility**: terms of service, `robots.txt`, and personal data; and proof that you can get one record.
- Details on assignment submission on Canvas: `https://canvas.colorado.edu/courses/143304/assignments/2852060`

The repository and write-up (December 7)

- **Reproducible**: a GitHub repository whose notebook or scripts collect the data and rerun the analysis, with a README.
- **A write-up** of about **2,000–3,000 words**, with figures.

The presentation (final week)

- About **5 minutes** plus questions: the question, the data, one finding.

Before you submit the proposal

- ☐ Is the question one the data can answer — and would anyone outside this class care?
- ☐ Does it name the exact source: an endpoint, a site's pages, or a set of PDFs?
- ☐ Did you retrieve **one real record**, and paste the request and the response?
- ☐ Did you read the terms of service and `robots.txt`? Does the data include people's names or other personal information, and how will you handle it?
- ☐ If the data changes over time, is your collector already running?
- ☐ Working in a pair? Ask the instructor before you submit.

The proposal is graded on a careful, good-faith plan, not on getting everything right: you may change your question or source later.

What comes next

Presentation, about 5 minutes (10%), in the final week, Nov. 30–Dec. 4. Repository and a 2,000–3,000-word write-up (25%), due Monday, December 7.

Late work

Loses 1% of its maximum value for each day after the deadline, unless you have an approved excuse.

Key takeaways

1. An API is a **documented contract**: structured, intentional access — clean JSON and stable endpoints, no reverse-engineering.
2. The skill transfers: **read the docs** → **build a parameterized request** → **parse nested JSON** one level at a time.
3. Always **paginate**: follow the `continue` token in a loop, or you silently keep only the first batch.
4. Be a good client: a descriptive **User-Agent**, serial requests, and `maxlag` for bulk jobs (the ethics of Ch. 2, applied). A client such as **mwclient** does the repetitive parts once you know what they are.
5. Know what your numbers count: a **pageview** is not a person, and its definition — and the bots behind it — have changed.
6. Openness is a *choice*, not a guarantee of completeness — Wikipedia's data still inherits its volunteer community's biases.

Next week: **Government APIs** — the Census, the Federal Reserve (FRED), and other public data services.